

Source-Traceable Research Intelligence: Evidence-Grounded Discovery, Question Answering, and Thesis-Extension Support

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Abstract—Publication lists expose outputs but do not answer full-text questions or help students formulate bounded, evidence-linked projects. We present an engineering case study of the TTLAB Research Intelligence Platform, whose central Thesis Extension Finder separates source facts, paper-stated future work, inferred gaps, and generated suggestions. The platform exposes corpus identity, provider metadata, evidence locators, uncertainty, and append-only review events. Historical v1 evaluation found strong lexical retrieval but weak citation targeting, answer completeness, abstention, and recommendation-feasibility evidence. A prospectively frozen remediation-v2 package is completed and validated and binds code, a frozen technical corpus, configurations, raw outputs, and a deterministic validator. Its AI-silver held-out results report answerability precision 66.7% (CI [44.4%, 90.0%]), exact-locator citation precision 9.1% ([0.0%, 18.8%]), and a Finder Hit@3 change of -20.0 pp ([-60.0 pp, +0.0 pp]). Known-positive topic recall is 75.0% ([45.5%, 100.0%]); OCR character error on a deterministic fixture is 0.0%. These metrics concern a technical only corpus; the public projection was exercised: no, and it may legitimately be empty until rights and review gates are satisfied. Source traceability means inspectable locators and provenance, not entailment or factual correctness. No recruited participants or independent human validation establish advisory usefulness, novelty, feasibility, or supervisor fit.

Index Terms—research intelligence, scholarly search, retrieval-augmented generation, recommendation, source traceability, reproducibility

I. INTRODUCTION

Large scholarly indexes connect papers and people [1]–[3], while recommender and expert-finding research supports content/profile matching [4], [5]. A laboratory archive has a narrower but less forgiving problem: missing PDFs, uncertain identities, generated advice, and incomplete review must remain visible instead of being hidden behind a polished search box.

The TTLAB precursor collected and displayed publications, generated lay summaries and notifications, and supported manual podcast preparation; evaluated full-text RAG was future work [6]. We extend it with full-paper ingestion, page-aware retrieval, citation-returning question answering, thesis-extension suggestions, topic/author exploration, and correction records. The Thesis Extension Finder is the central product

contribution. It ranks evidence-bearing papers before adding an explicitly generated project proposal; it cannot establish novelty, data access, workload, or researcher availability.

Here *source-traceable* means that an output retains paper, chunk, page/section, provider/model, time, warning, and review-state locators. Traceability is not entailment, completeness, factual correctness, feasibility, or endorsement. This distinction governs both the application and the claims below.

Three questions organize the study: **RQ1**, can a heterogeneous archive become an internally consistent corpus and how do retrieval modes compare; **RQ2**, how strong are evidence-linked answers, extension suggestions, topics, and author links under their stated constructs; and **RQ3**, how reproducible, reviewable, and responsibly bounded is the artefact? Contributions are a hash-bound corpus/index contract; a modular, evidence-first platform; a prospective source-derived protocol plus preserved negative historical evidence; and a sanitized reproduction boundary. No recruited participants were used, so usefulness and supervisor suitability remain untested.

II. RELATED WORK

Scholarly discovery systems progressed from autonomous citation indexing to literature/author graphs and open entity catalogues [1]–[3]. Their scale is useful for discovery, but a small institutional system must also expose why a paper was included, which file supplied a passage, whether an identity is confirmed, and whether generated material has been reviewed. Research-paper recommendation surveys likewise show that methods, user models, and evaluation protocols vary widely [4]; an evidence match is not the same as a feasible supervisory recommendation. Expert-finding models infer expertise from documents [5], so our author links are deliberately phrased as publication-derived associations rather than availability or endorsement.

BM25-style lexical ranking remains a strong baseline [7]. Feature hashing is a memory-bounded sparse representation, not a learned semantic encoder [8]. Sentence-BERT and DPR provide learned sentence/passage representations [9], [10]; BEIR cautions that rankings vary across tasks and domains

TABLE I
HISTORICAL V1 CORPUS AND INDEX BOUNDARY.

Catalogue	134	Local PDFs	98
Eligible papers	96	Needs text/review	36
Raw chunks	735	Eligible chunks	719
PDF mismatches	2	Unknown sections	199

[11]. We therefore compare modes rather than presuming dense or hybrid superiority.

RAG conditions generation on retrieved evidence [12], and later work adds retrieval/self-critique controls [13]. Evaluation frameworks inspect context relevance and faithfulness [14], [15]; atomic decomposition exposes claim-level support [16]. Returned citations nevertheless can target the wrong passage, and high support can coexist with missing answer points. Topic interpretability likewise requires inspection rather than relying on aggregate coherence [17]. Human-in-the-loop guidance motivates visible uncertainty and correctable outputs [18].

III. METHODOLOGY AND SYSTEM

A. Units, Corpus, and Evidence Boundary

The study combines design-science-oriented artefact construction with controlled offline evaluation. Units are papers/chunks for corpus integrity; queries for retrieval; atomic claims, citation links, and expected answer points for QA; synthetic profiles and ranked paper sets for recommendation; papers/labels for topics; and executable stages for performance. TTLAB authorized the project and corpus use, but that is neither ethics-board approval nor blanket third-party redistribution permission.

Design decisions were mapped to observable failures. Deterministic import and hash-checked indexes address silent corpus drift; page locators and typed evidence address unverifiable generation; evidence-only Finder output addresses users who do not want generated advice; append-only review events address untraceable correction; and explicit unavailable states address missing providers or PDFs. Each mechanism is evaluated at the narrowest valid unit instead of treating a passing user-interface test as retrieval evidence or a citation count as faithfulness.

The historical v1 snapshot starts with 134 records and 98 available PDFs. A paper enters technical experiments only when metadata, a permitted PDF, extracted text, and title/content checks agree. Table I shows that boundary: 96 papers and 719 of 735 chunks were eligible; 36 records needed text/review, 2 PDFs were mismatches, and 199 eligible chunks retained an Unknown section instead of a forced label. A 40-case AI-silver check measured 0.900 section accuracy.

Keyword FTS, 256-dimensional signed feature hashing, and the pinned 384-dimensional `all-MiniLM-L6-v2` encoder share an ordered eligible-chunk boundary. Feature hashing is not labelled learned semantic retrieval. Manifests bind corpus/configuration hashes, model revision, dimensions, code, ordered source hashes, and completeness; startup and experiments reject stale or partial indexes. Historical artefacts retain

their own execution identities instead of being relabelled as current.

Remediation-v2 is separately identity-bound to source commit `0d4b9bdcb657` and snapshot `corpus-f4638c633bea82b0`. Its technical boundary contains 96 papers and 719 chunks. The package status is completed and validated. These generated values prevent a manuscript build from silently combining a new corpus with old results.

B. Evaluation Protocol

Historical v1 used 50 source-derived retrieval cases, one AI process for source-linked labels, a later shuffled source-first reinspection without rankings, and a fixed 30/20 development/test split. On answerable queries, set recall is $|G_q \cap P_{q,k}|/|G_q|$; MRR and nDCG@10 complement it. Query-bootstrap intervals and Holm-corrected paired tests were reported, but correlated AI-label error was not modelled.

The prospective remediation protocol freezes its corpus, source-derived cases, locators, splits, configuration, environment lock, and output schema before execution. Cases were constructed from sources without consulting rankings; held-out outputs are immutable after the run. Ask uses fixed keyword retrieval, a deterministic extractive answerer, and a predeclared answerability rule over 12 test cases. Claims are decomposed and scored for exact evidence locators, citation completeness, answer-point coverage, and abstention. Finder compares evidence-only ranking with full, separately labelled suggestions on the same retrieval response for 5 held-out synthetic profiles. Profiles are proxies, not students.

Topics use 8 source-derived, known-positive test cases. Because labels are not exhaustive, the protocol reports positive-label recall, coverage, and unadjudicated predictions—not precision, F1, or exact match. A deterministic raster fixture exercises the OCR path but is not a corpus OCR estimate. Cluster-level bootstrap intervals quantify sampling of cases, profiles, or papers only. A shuffled second AI pass records disagreements; repeated AI inspection is not independent human validation, inter-rater reliability, or entailment assessment.

Figures 1 and 2 summarize the implementation. The React interface provides papers, search, Ask, Finder, topic/author, evaluation, and protected review routes. It renders freshness, provider, warning, review state, citations, and evidence-only alternatives, including explicit empty, unavailable, and retry states. FastAPI separates public reads and transient generation from authenticated mutations; exact origins/hosts, bounded requests, safe file/URL handling, redacted logs, and append-only attributed review events are enforced. Profile inputs are non-persistent by default.

Page-aware extraction distinguishes blank, scanned, and mixed pages. Search returns mode, score components, pages, sections, freshness, and the active corpus identity. Runtime grounding checks locator presence and weak lexical overlap, not entailment. The answerer extracts bounded sentences. Finder first constructs source-backed baseline work, paper-stated future work, and an explicitly inferred gap; only then

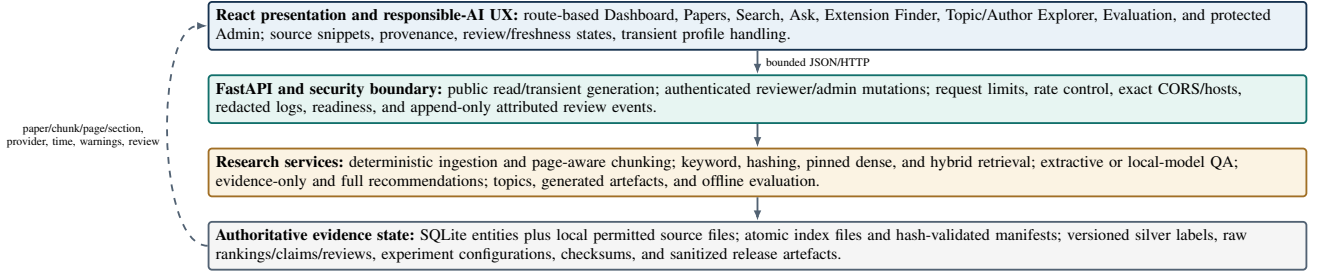


Fig. 1. Implemented architecture. Administrative mutations cross an authenticated boundary; evidence locators return through every layer.

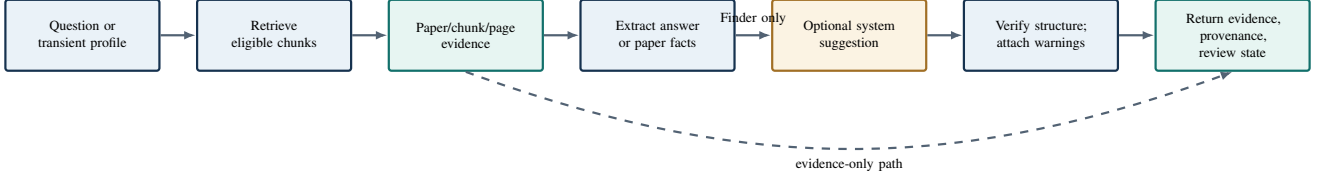


Fig. 2. Unified Ask/Finder evidence flow. Evidence-only output and system suggestions remain distinct.

may it add a labelled suggestion with assumptions, MVP, stretch scope, skills, data, risk, and evaluation.

Technical eligibility is not publication approval. Public routes use a stricter projection requiring rights clearance, review, publication state, and an aligned public index generation; they never substitute the technical corpus. Consequently the public projection at the audited/remediated snapshot may be empty. The remediation evaluation uses only the technical boundary and is not evidence of public behavior. The optional Ollama provider did not contribute historical v1 results, so no response, digest, or latency comparison was invented.

C. Evidence Contracts and Failure Semantics

Derived state is generation-addressed. Extraction records bind exact PDF bytes; chunk records bind the extraction generation and exact UTF-8 text; each authoritative index manifest binds the ordered eligible source identities and active configuration. Reconciliation validates that chain before rebuilding. Index files are written as immutable generations, and the active pointer changes only after database status is committed. A bounded demo build cannot target the authoritative index. Startup and requests fail closed on a missing, partial, stale, or differently scoped manifest rather than silently falling back to another corpus.

Ask decomposes an answer into checkable claims, retains only citations used by those claims, and exposes an unsupported or abstained state when fixed answerability evidence is insufficient. A citation establishes a locator, not that the cited passage entails the prose. Finder ranks evidence-bearing papers before constructing suggestions; candidate facts are recomputed from the selected paper, while unresolved time, data, novelty, and feasibility remain unknown or assumptions. The public API omits technical hashes and local paths but retains safe paper/page/chunk navigation.

Review state is likewise typed. Corrections, review decisions, and the effective display projection remain separate;

TABLE II
EVIDENCE CONTRACTS AND PROHIBITED INFERENCES.

Surface	Evidence retained	Does not establish
Search	corpus, chunk, rank, mode	relevance truth
Ask	claim, locator, warning	entailment/correctness
Finder	fact, gap type, suggestion	novelty/feasibility
Topics/authors	source links, review state	expertise/availability
Review	actor type, event chain	human approval by AI

TABLE III
HISTORICAL V1 HELD-OUT RETRIEVAL (19 ANSWERABLE QUERIES).

Mode	R@3	R@10	MRR	nDCG@10
Keyword	0.939	1.000	0.947	0.958
Feature hashing	0.654	0.704	0.588	0.582
Learned dense	0.916	0.989	0.939	0.939
Heuristic hybrid	0.795	0.969	0.822	0.848
Dev-tuned hybrid	0.909	0.978	0.912	0.924

events are attributed and hash chained. An AI actor can record an AI-review outcome but cannot create a human approval. Provider-offline, stale-index, and empty-public-corpus states are first-class responses with retry or evidence-only paths. Table II summarizes the permitted inference boundary.

IV. RESULTS

A. RQ1: Historical v1 Corpus and Retrieval

All three historical v1 indexes covered 719/719 eligible chunks; excluded/raw chunks could not enter that evaluation. Author cleanup removed navigation artefacts and repaired punctuation aliases, while 13 possible same-person pairs remain unmerged pending external confirmation. Missing meta-data stays empty.

Table III shows that keyword had the highest historical point estimates: Recall@10 1.000 (CI [1.000, 1.000]) and MRR 0.947 ([0.868, 1.000]). Dense MRR was 0.939 ([0.842,

TABLE IV
HISTORICAL V1 NEGATIVE AND NULL EVIDENCE.

Outcome	Estimate/count	CI or denominator
Citation correctness	0.653	[0.592, 0.715]
Strict answer-point coverage	0.1358	11/81; [0.067, 0.211]
Unanswerable abstention	0.250	1/4; [0.000, 1.000]
Finder relevance change	-0.036	[-0.119, 0.048]
Finder full-mode return	67/84	0.798; 17 missing

TABLE V
PROSPECTIVE HELD-OUT TECHNICAL OUTCOMES (GENERATED MACROS).

Outcome	Estimate	Cluster CI
QA cases	12	–
Answerability precision	66.7%	[44.4%, 90.0%]
Exact-locator citation precision	9.1%	[0.0%, 18.8%]
Strict answer-point coverage	18.8%	[6.2%, 30.0%]
Finder profiles	5	–
Finder Hit@3 change	-20.0 pp	[-60.0 pp, +0.0 pp]
Topic cases	8	–
Known-positive topic recall	75.0%	[45.5%, 100.0%]
OCR fixture character error	0.0%	–

1.000]); tuned-hybrid MRR was 0.912 ([0.816, 1.000]). No corrected comparison established hybrid superiority, and equivalence was not tested. Removing all 6 heuristics recovered Recall@10 1.000 and MRR 0.947. Every mode answered the single held-out unanswerable query, an unstable but adverse signal.

B. RQ2: Historical Diagnosis and Prospective Evidence

Table IV summarizes the principal historical negatives. Historical v1 classified 331/3/0 of 334 extractive claims as supported/partial/unsupported, yet strict answer-point coverage was only 0.1358 (11/81), and just 1/4 unanswerable cases abstained (49/50 cases answered overall). Its case-bootstrap intervals exclude AI-label uncertainty. Finder evidence-only mode returned 84/84 items, whereas full mode returned 67/84, leaving 17 cutoff slots scored as zero; the relevance change was negative and its interval spanned zero. Conditional on the 67 returns, source fidelity passed 67/67; pass/partial/fail counts were feasibility 0/0/67, MVP 64/3/0, and evaluation plan 31/7/29; separation passed/failed 66/1, while usefulness failed 67/67. Feasibility failure records unverified data, time, and skills, not known infeasibility. Topic silver evidence bound through 119 exact chunk IDs and 1 deterministic same-paper locator rebound; 1 snapshot-drift binding preserved the persisted excerpt exactly. Binding integrity does not establish label correctness. These are AI-reviewed silver findings.

Table V is populated only from the validated package. Ask metrics test the fixed answerability gate and exact frozen locators; they do not establish factual correctness or entailment. Comparison of historical strict coverage 0.1358 ([0.067, 0.211]) with 18.8% ([6.2%, 30.0%]) is descriptive and non-paired because the protocols use distinct frozen cases. The paired Finder delta is reported without a superiority claim, while template checks establish a data-contract property rather than project usefulness. Topic recall applies only to listed known positives; other predictions remain unadjudicated. OCR

TABLE VI
SELECTED LOCAL STAGE MEDIANS (S) AND PEAK RSS (MiB).

Stage	Cold	Warm	RSS
Discovery	2.952	0.760	65.1
PDF extraction	8.974	5.357	123.1
Dense index	153.033	75.343	737.4
Hybrid retrieval	83.195	9.084	599.3
Offline answer	83.628	9.052	598.1
Rendered routes	13.732	13.307	361.1

character error applies only to the deterministic fixture. All are AI-silver technical-corpus judgments without human validation. Author links remain publication-derived associations, unresolved identities remain separate, and AI review events cannot confer human approval.

C. RQ3: Reproduction and Operational Boundary

The historical v1 local profile measured 17 stages with process-cold and warm repetitions at concurrency one and 0 recorded failures. “Cold” means a fresh process; OS caches were not flushed, and this is neither load nor capacity evidence. An external harness reacquired 3 pinned CC BY JATS/XML papers and obtained 3/3 fixed lexical top-one matches; that establishes format compatibility and trivial discrimination only, not cross-domain quality or PDF ingestion.

Table VI selects representative stages; omitted repetitions remain in the hashed raw record. Network discovery dominates variability, dense indexing dominates local compute/memory, and warm/cold differences are workload-specific. No concurrency sweep, cache flush, or saturation test was conducted. The prospective package records source commit 0d4b9bdc657, corpus snapshot corpus-f4638c633bea82b0, raw outputs, configuration, environment lock, and post-run validation attestation. Reproduction rebuilds allowed derived state, runs tests/evaluations, compiles manuscripts, preflights PDFs, creates a restricted-content-free bundle, and emits checksums. Exact full-corpus reruns still require lawful access to local PDFs and the pinned dense model.

V. DISCUSSION, THREATS, AND ETHICS

The principal result is an evidence boundary, not a new ranking algorithm. Historical v1 showed that complete indexing and extractive support could coexist with weak citation choice, answer completeness, abstention, and Finder feasibility. The prospective measures narrow those constructs to a fixed answerability gate, exact source locators, paired recommendation ranks, known-positive topics, and a pipeline fixture. They do not erase the historical failures or convert contract conformance into benefit. Human studies must test usefulness, novelty, workload, and trust.

A. Evidence-Driven Design Decisions

The historical retrieval evidence makes keyword the defensible default for that snapshot: it had the strongest held-out point estimates and simple failure semantics. Dense search remains an explicit paraphrase-oriented alternative, not a claim

of semantic quality. The tuned hybrid remains experimental because neither paired analysis nor ablations justified superiority. A future router must earn its complexity on a new locked test set.

QA therefore needs a multi-dimensional result state. A response can be source-derived but answer the wrong intent, cite an off-target passage, omit expected points, or answer when evidence is insufficient. The interface separates answerability, claim support, citation use, and completeness, and does not project aggregate scores onto a live answer. If the fixed gate fails, ranked evidence or an explicit unsupported state is safer than composed text.

Finder needs the analogous boundary. Ranking a paper can be justified by passages, interests, and stated constraints; proposing a thesis adds novelty, access, workload, skills, and evaluation assumptions the corpus cannot settle. The record distinguishes baseline work, paper-stated future work, inferred gap, and system suggestion. Evidence-only mode bypasses suggestions; full mode exposes assumptions and risks rather than a hidden suitability score. The historical feasibility failures and unfilled ranks motivate a supervisor checklist, not more assertive prose. No recommendation is supervisor-approved until an authorized human records that state.

Administrative review is an append-only evidence event rather than a silent overwrite. Events bind actor type, prior/new states, rationale, time, target hash, and chain hash. AI actors can mark AI-reviewed or needs-reprocess states but cannot impersonate human approval. Historical answers that cannot be exactly regenerated retain that limitation, and ambiguous identities remain unmerged.

The external smoke test and performance profile answer deployment questions only narrowly. The former shows that three licensed XML documents can enter the chunk contract and be separated by obvious lexical queries; it does not validate another discipline, language, or PDF path. The latter identifies expensive stages for a single local operator but offers no arrival-rate, tail-latency-under-load, or capacity evidence. Production planning must add representative traffic, concurrent users, failure injection, monitoring, and recovery tests without reusing these single-user medians as service-level objectives.

Construct validity is limited by AI-silver labels, source-derived positive-only topics, synthetic-profile proxies, and a fixture rather than corpus OCR. One AI procedure performs both shuffled passes, so correlated error remains and intervals exclude label uncertainty. Historical internal validity is limited by small test sets, one retrieval unanswerable, descriptive ablations, and a deterministic answerer; prospective cases remain small and source-derived. External validity is limited to one English-language laboratory snapshot and local hardware. Reproducibility remains bounded by lawful PDF/model access, network discovery, and OS caches.

TTLAB authorized project and corpus use. No participants were recruited and no ethics approval or exemption is claimed. Authorization does not transfer redistribution rights: PDFs, live database, derived text, secrets, model caches, and private prompts are excluded from the sanitized release.

Public deployment additionally requires authenticated review, controlled origins, rate limits, monitoring, correction paths, backups, and accessibility validation. Datasheet, model-card, and risk-management practices inform disclosure [19]–[21]. Codex/LLM assistance covered code, silver-label preparation, analysis, and editing; it was not a human assessor. The current Tectonic PDF is intentionally not claimed as tagged PDF or PDF/UA; figure descriptions remain in source for a capable venue toolchain.

VI. AVAILABILITY AND CONCLUSION

The sanitized bundle contains code, permitted metadata, sanitized experimental records, configurations, figure sources, manifests, hashes, and reproduction instructions. It excludes restricted content and is not the publisher package. Historical v1 negative and null findings remain alongside the completed and validated prospective evidence rather than being overwritten. The newer results apply to a technical only corpus, with public projection exercised: no. Technical evaluation is therefore not public behavior. The artefact is a source-traceable research-intelligence prototype, not a validated academic advisor: human utility, identity decisions, PDF rights, deployment controls, and venue compliance remain external validations.

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